Example 3: Consider the following 2D array named temperatures:

	0	1	2	3	4	5	6
0	12.5	12.2	14.0	13.0	13.2	13.0	12.2
1	**15.2**	**11.9**	**11.8**	**13.2**	**12.7**	**14.5**	**12.4**
2	9.75	11.8	11.9	13.5	12.7	14.7	13.0
3	14.0	13.2	11.8	14.0	14.3	11.0	11.9

Perform the following:

1. Print out the value of highlighted indexes.
2. Print out the value of highlighted row.
3. Find out the maximum, minimum and average values.

```
public class Array2DExample3 {
     public static void main(String args[]) {
     double[][] temperatures = {{12.5,12.2,14.0,13.0,13.2,13.0,12.2},
                                {15.2,11.9,11.8,13.2,12.7,14.5,12.4},
                                {9.75,11.8,11.9,13.5,12.7,14.7,13.0},
                                {14.0,13.2,11.8,14.0,14.3,11.0,11.9}};
     double min,max,sum;
     min= temperatures[0][0]; // Assume the first item is the minimum
     max= temperatures[0][0]; // Assume the first item is the maximum
     sum=0;
     System.out.println(temperatures[1][5]);

     for (int col=0; col< temperatures[1].length; col++)
          System.out.print(temperatures[1][col] + " ");
     System.out.println();

     for (int row=0; row< temperatures.length; row++)
     {
        for (int col=0; col< temperatures[row].length; col++)
          {
            if (min > temperatures[row][col])
                    min= temperatures[row][col];
            if (max < temperatures[row][col])
                    max= temperatures[row][col];
               sum=sum+ temperatures[row][col];
          }
     }
     System.out.println("The minimum value is " + min);
     System.out.println("The maximum value is " + max);
     System.out.println("The average is " + sum/28);
   }
}
```

```
Markers   Properties   Servers   Data Source
<terminated> Array2DExample3 [Java Application] /Library
14.5
15.2 11.9 11.8 13.2 12.7 14.5 12.4
The minimum value is 9.75
The maximum value is 15.2
The average is 12.833928571428572
```

Example 4: Write down a Java program to read a 2-dimensional array x[4][4] then perform each of the following:

1. Find the maximum and minimum in the third row
2. Replace the two diagonals with 1

```
import java.util.Scanner;
public class Array2DExample4 {
     public static void main(String args[]) {
    int [][] x = new int[4][4];
    int min,max;
    Scanner s = new Scanner(System.in);
    System.out.print("Enter the array elements: ");
    for (int i=0; i< 4; i++)
          for (int j=0; j< 4; j++)
          x[i][j] = s.nextInt();
    s.close();

    for (int i=0; i< 4; i++)
    {
        for (int j=0; j< 4; j++)
                System.out.print(x[i][j] +" ");
     System.out.println();
     }
    min= x[3][0]; // Assume the first item is the minimum
    max= x[3][0]; // Assume the first item is the maximum

    for (int j=0; j< 4; j++)
    {
       if (min > x[3][j])
          min= x[3][j];
       if (max < x[3][j])
          max= x[3][j];
     }
     System.out.println("The minimum value is " + min);
     System.out.println("The maximum value is " + max);

     for (int i=0; i< 4; i++)
     for (int j=0; j< 4; j++)
     {
                if (i == j) // Condition for principal diagonal
                     x[i][j] = 1;
                if ((i+j) == 3)// Condition for secondary diagonal
                     x[i][j] = 1;
     }
     for (int i=0; i< 4; i++)
     {
     for (int j=0; j< 4; j++)
                System.out.print(x[i][j] +" ");
     System.out.println();
     }
  }
}
```

```
Markers  Properties  Servers  Data Source Explorer  Snippets  Console
<terminated> Array2DExample4 [Java Application] /Library/Java/JavaVirtualMachines/jdk1.8.0_112.jdk/
Enter the array elements: 22 12 23 24 44 13 32 34 55 18 17 90 12 14 78 55
22 12 23 24
44 13 32 34
55 18 17 90
12 14 78 55
The minimum value is 12
The maximum value is 78
1 12 23 1
44 1 1 34
55 1 1 90
1 14 78 1
```

Homework:

Q1> Write a Java program to count the number of occurrences of an element in an 2D array.

Q2> Write a Java program to print negative numbers in an 2D array.

Q3> Write a Java program to read of an 2D array then to print the two diagonals elements only.

Q4: write java program to print elements under and over the main diagonal in 4×4 2D Array.